Yongfeng Xu(Justin Holland)

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EDUCATION

Nanyang Technological University 08/2025-Present
- Expect Master of Science
- Major: Communications Engineering
- GPA: 4.1/5.0; Score: NA/100

South China University of Technology 09/2021-06/2025
- Bachelor of Engineering
- Major: Information Engineering
- GPA: 3.44/4.0; Score: 84.4/100
- Completed *Data Analysis with R Programming/ Google Data Analytics* on Coursera.

Google Data Analytics@Coursera 08/2024-12/2024
- Major: Data Analysis with R Programming
- Completed *Data Analysis with R Programming/ Google Data Analytics* on Coursera.

INTERNSHIP EXPERIENCE

China Telecom 06/2024-07/2024
Wireless Network Engineer
- Participated in basic operations and deployment of unmanned aerial vehicles (UAVs).
- Engaged in fiber optic fusion splicing activities.
- Performed VSAT (Very Small Aperture Terminal) ground reception tests.

RESEARCH EXPERIENCES

UWB Radar Car Scene Application 08/2024-12/2024
- Utilized Python (Pandas, NumPy) for data cleaning and preprocessing to remove noise and enhance data quality.
- Developed algorithms in Python and MATLAB, employing Scikit-learn for parameter optimization to detect living beings within vehicles.
- Embedded software programmed in C and C++ to ensure real-time algorithm performance on hardware.
- Conducted simulation tests in MATLAB and field tests with radar hardware on actual vehicles to validate algorithmic accuracy.

Design of Intelligent Health Aid Agricultural Applet 03/2024-06/2024
- Designed a user-friendly front-end interface using HTML, CSS, and JavaScript.
- Created a back-end database with SQLite for product data storage and retrieval.
- Integrated an AI module using R and TensorFlow to enable health question-answering functionality. Linked the front and back ends through API connections to facilitate interactive, intelligent responses.

Research on New Radio Direction Finding System Based on Scattering Imaging Principle 02/2023-02/2024
- Simulated and optimized antenna array layout and performance with MATLAB.
- Analyzed scattering characteristics using HFSS.
- Developed algorithms in C++, incorporating directional optimization methods such as beamforming to enhance directional accuracy.
- Established communication between the antenna array, receiver, and host system through I2C protocol, ensuring seamless module integration.

Frequency Modulation Communication System Based on Phase-Locked Loop 12/2023-02/2024
- Simulated the signal transmission and reception circuits in Multisim and optimized the design.
- Selected and soldered components like PLLs and amplifiers, followed by comprehensive testing.

- Used MATLAB and Simulink for system-wide simulations, assessing signal transmission quality and robustness, and subsequently optimized the circuit design. Finalized the circuit board layout in Cadence for printing and implementation.

MCU Design of Electronic Calendar Clock 06/2023-09/2023

- Designed and created circuit diagrams using Proteus to ensure accurate component connections.
- Programmed the microcontroller with C language to enable time display, calibration, and adjustment functions.
- Developed a key debounce algorithm to enhance input accuracy and system response speed.
- Write a design report for electronic clock design.

PATENT

Synchronous Demodulation Method and System for Passive RFID Tags 10/2025

- China National Intellectual Property Administration
- No. ZL 2025 1 0060679.1

EXTRACURRICULAR ACTIVITIES

Campus UP 08/2025-Present

- Organize the weekly event about the club.
- Connect to potential sponsors.

Radio Association 10/2021-01/2023

- Organized the Wireless Cup competition and assisted with technical sharing sessions to advance radio technology.
- Collaborated to support educational and competitive initiatives, and encouraged participation in radio-related activities, advancing technical skills through competitions and workshops.

English Corner 03/2023-07/2024

- Organized regular English-speaking practice sessions
- Involved in planning and hosting large-scale events, and liaised with other associations, such as the International Students Association to create a welcoming environment for language and cultural exchange.

HONOURS AND AWARDS

- Successful Participant in Mathematical Contest in Modeling 05/2024
- National Undergraduate Training Program for Innovation and Entrepreneurship Finalization 04/2024
- Merit Award in the Greater Bay Area Cup Guangdong-Hong Kong-Macao Financial Mathematical Modeling Competition 11/2023
- First Prize in the National Undergraduate Innovation and Entrepreneurship Competition (awarded to the top 5%) 07/2023
- Third Prize Scholarship (awarded to the top 15%) 12/2022

SKILLS AND INTERESTS

- Programming languages and development tools: C++, MATLAB, Python, R, SQL.
- Software and simulation tools: Multisim, Quartus, Global Mapper, Mathtype, VMware, Git, Photoshop.
- Measurement and testing equipment: oscilloscope, spectrum analyzer, function generator, multimeter, signal generator, LCR meter, logic analyzer, frequency counter, network analyzer.
- Interests and hobbies: excursion, calligraphy, football, chess, swimming.

SELF-DESCRIPTION

- I am eager to learn and continuously improve my skills, with strong adaptability and problem-solving abilities. I take responsibility in my work, approach tasks with dedication, and strive to deliver high-quality results. With an open and proactive personality, I enjoy communicating and collaborating with others, and I value teamwork as a key to achieving success.